

Protocol followed for DICOM analysis in 3D Slicer

Preparation of the DICOM model

- 1) Open 3D Slicer
 - a) Open Dicom – Import DICOM files
 - b) for NMDID choose THIN_BONE_HEAD
 - c) Volume Rendering – click eye icon
 - d) Centre Volume
 - e) Display – Select Preset
 - i) NMDID – CT (Computer Tomography) Chest contrast enhanced
- 2) Alignment of Model
 - a) Allocate 3 of the Frankfort horizontal plane landmarks
 - b) Open FHP_landmarks.mrk.json
 - c) FHP_landmarks.mrk.json" (dragged into the open Markups module)
 - d) Landmark names NEED to be consistent
 - i) poR - right porion
 - ii) poL – left porion
 - iii) zyoL – left orbitale
 - e) Open python console, copy + paste code '[001_code for fhp alignment](#)'
 - f) Open (dragged into the open Markups module) and allocate 4 landmarks to establish the Frankfort horizontal plane
 - i) FH4_landmarks.mrk.json"
 - g) Adjust Plane Settings > Type – Plane fit
 - h) Leave plane name 'P'
- 3) Allocate hard tissue landmarks
 - a) Open via hard tissue.mrk.json
- 4) Open and run '[002_axes and mastoid length](#)' code
 - a) The second points of Mastoid length R and L need to be randomly manually allocated then adjusted via the '[003_code for orthogonal projection](#)'
 - b) Then make 'P' visible and allocate the second points of all the axes manually according to the following description:

main axis of right mastoid L	general axis passing through the central aspect of the right mastoid process in lateral view from maiR
main axis of left mastoid L	general axis passing through the central aspect of the left mastoid process in lateral view from maiL
main axis of right mastoid I	general axis passing through the central aspect of the right mastoid process in an inferior view from maiR
main axis of left mastoid I	general axis passing through the central aspect of the left mastoid process in an inferior view from maiL
mandibular ramus axis R	axis connecting the most superior part of the right condylar process and the right gonion of the mandible
mandibular ramus axis L	axis connecting the most superior part of the left condylar process and the left gonion of the mandible

5) Allocate soft tissue landmarks

- a) Open via **soft tissue.mrk.json** (dragged into the open Markups module)
- b) Toggle volume rendering bar for visibility of skin

6) Add the other planes

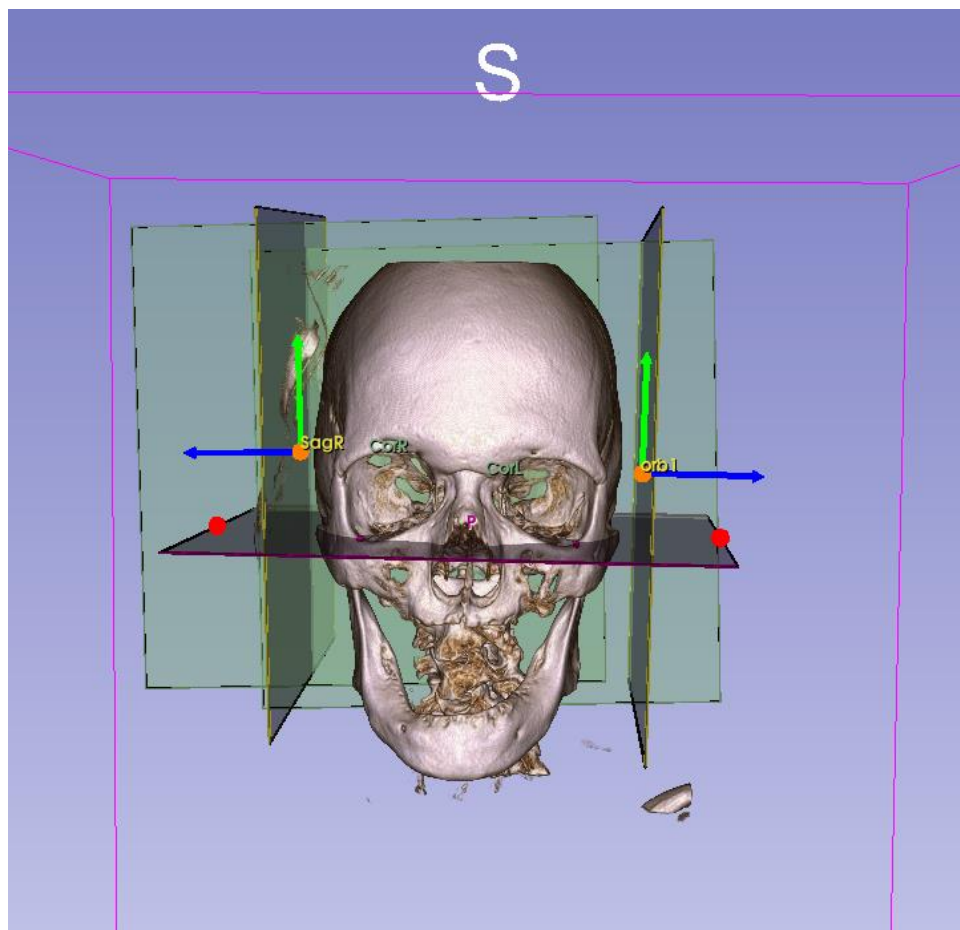
- a) Duplicate the FHP ONCE (right click on Markups list > Clone)
- b) Rename it to SagR
 - i) Clone SagR 3 times and rename them SagL, CorL, CorR
 - ii) Time to rotate the new planes
 - (1) Open Transforms
 - (2) Click SagR, SagL, CorR, CorL then the >O arrow
 - (3) Adjust Rotation LR to 90 degrees
 - (4) Click again on all of them and on 'Harden Transform'
 - (5) Click on CorR and CorL, apply the transform IS 90, harden
 - iii) Adjust the planes to meet the dots described
 - (1) Toggle the visibility of obsR, obsL (soft) and the manual posterior point of both main axes of mastoid I (called something like "main axis of left mastoid I-6")

(2) Then enable under `Display~` `Interaction` Visibility` `enable translation` for all and manually toggle until they cross the landmarks described above

(a) If they don't move, unlock their control points!

SagR			soft	obsR
SagL			soft	obsL
CorR			hard	manual endpoint of main axis of right mastoid I
CorL			hard	manual endpoint of main axis of left mastoid I

After all, it should look like this:



1) Linear Measurements

a. Programmable linear measurements

i.

measurement	definition	direction	Tissue type
ear height L	saR-sbaR	bilateral	soft
ear height R	saL-sbaL	bilateral	soft
ear width R	praR-paR	bilateral	soft
ear width L	praL-paL	bilateral	soft
ear insertion height R	obsR-obiR	bilateral	soft
ear insertion height L	obsL-obiL	bilateral	soft
nasal dorsum length	se-prn	midline	soft
soft nose height 1 se-sn	se-sn	midline	soft
soft nose height 2 g`sn	g`sn	midline	soft
hard nose height 1 (n-ss)	n-ss	midline	hard
hard nose height 2 (g-ss)	g-ss	midline	hard
facial height	n-gn	midline	hard
ear protrusion R obs-hx	obsR-hxR	bilateral	soft
ear protrusion L obs-hx	obsL-hxL	bilateral	soft
for hard nose angle	n-rhi	midline	soft

Generate these using the script [004 code for linear measurements](#)

Manual Linear distances

- iv) Don't forget to adjust the hard nose angle line to meet the FHP
- v) enable under `Display~` Interaction` Visibility` `enable translation` for all and manually toggle until it crosses 'P'

Saving linear data

Use the code [data to excel-linear](#)

Use Ctrl+M to copy linear measurements to clipboard (if everything worked, it's 23 measurements, you can delete the axes, ear protrusions and the for hard nose angle as they are not needed – 14 remains)

7) Second round of programmed lines

The run the code [005 Generating intersection points](#)

Mastoid lateral angle -needs to be running mediolaterally along the coronal plane to be precise with measurement

Instead of the projected lines, the intersection points **MLA R** and **MLA L** will form the reference line and extend the line on both sides via toggling

Open empty name 'for L mastoid angle')

Populate the line with MLA R and MLA L as endpoints

Clone the line and rename it to 'for R mastoid angle'

Unlock the endpoints on both lines

Go to 'Display'>Interaction handles> Enable translation > Visibility

Adjust them to be on their respective sides

Now, the scripts for projected lines can be run from [006 code to generate ALL PROJECTED LINES](#)

Now run the code for prepopulating angles: [007 prepopulating 2 points for angle](#)

Allocate manually the third points according to the table:

Angle	A (coded)	Vertex (coded)	B (manual)	B name example
ear angle R	sbaR	EA R intersection	anterior end of ear height R projected	Ear heigh R projected - 2
ear angle L	sbaL	EA L intersection	anterior end of ear height L projected	Ear height L projected 2
ear insertion angle R	obiR	EIA R intersection	anterior end of ear insertion height R projected	ear insertion height R projected-2

ear insertion angle L	obiL	EIA L intersection	anterior end of ear insertion height L projected	ear insertion height L projected-2
ear protrusion R	hxR	obsR	posterior end of ear protrusion R obs-hx projected	ear protrusion R obs- hx projected-2
ear protrusion L	hxL	obsL	posterior end of ear protrusion L obs-hx projected	
mastoid anterior angle R	maiR	MAA R intersection	Anterior end of main axis of right mastoid L projected from lateral view	main axis of left mastoid L projected-1
mastoid anterior angle L	maiL	MAA L intersection	Anterior end of main axis of left mastoid L projected	
mastoid lateral angle R	maiR	MLA R intersection	Lateral end of for R mastoid lateral angle projected	for R mastoid anterior angle projected-2
mastoid lateral angle L	maiL	MLA L intersection	lateral end of for L mastoid lateral angle projected	for L mastoid anterior angle projected-1
mandible ramus angle R	goR	MRA R intersection	medial end of mandibular ramus axis R projected	
mandible ramus angle L	goL	MRA L intersection	medial end of mandible ramus axis L projected	mandibular ramus axis R projected-2
nasal root angle	prn	NRA intersection	anterior end of nasal dorsum length projected	nasal dorsum length projected-2
hard nose angle	anterior end of toggled line for	HNA intersection	anterior end for hard nose angle projected	

	hard nose angle			
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You can now run script: [angles to excel](#)

Once done, run the command to copy all angle measurements to clipboard via CTRL+T

14 angles should be in total